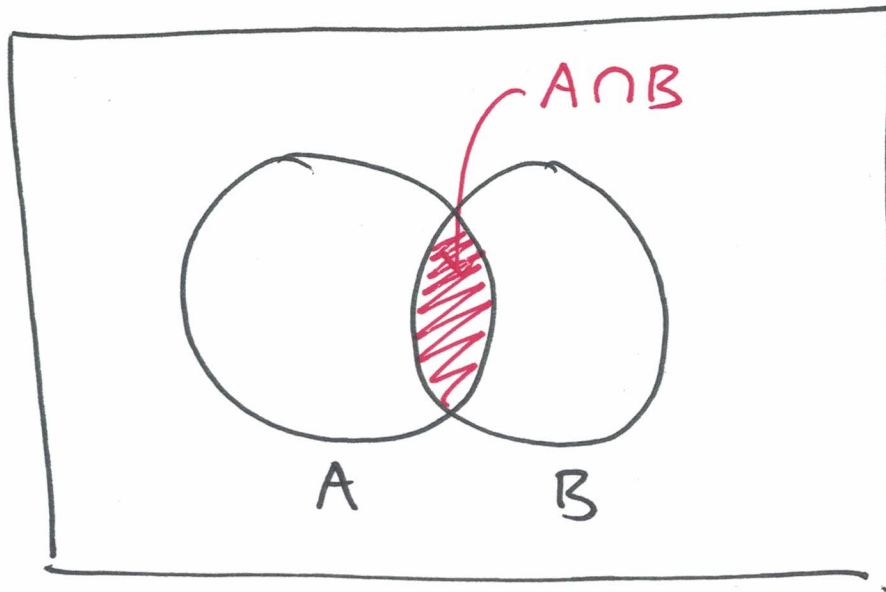


CONDITIONAL PROBABILITY

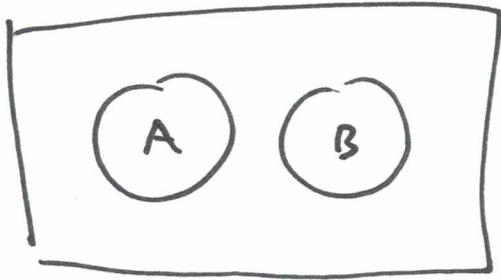
$P(A | B)$

" PROBABILITY OF A
GIVEN B "

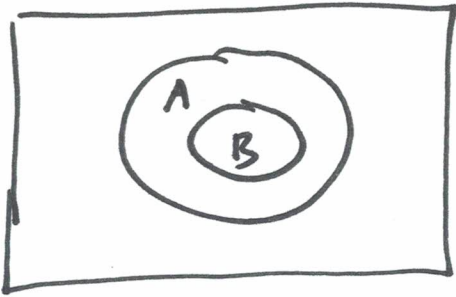


$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

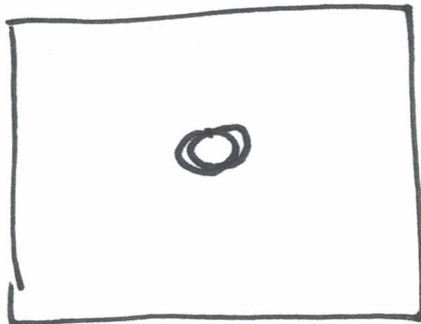
COULD GIVE A LOT OF INFORMATION:



$$P(A|B) = 0$$

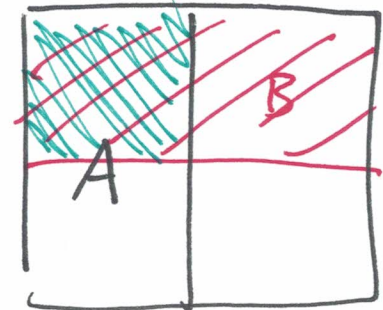


$$P(A|B) = 1$$



$P(A)$, $P(B)$ VERY SMALL

$P(A|B)$ BIG



$\begin{matrix} \diagup & x \\ y & \diagdown \end{matrix}$	0	1	2	3	4	5	6	7	8	9
+1	10	2	3	4	5	6	7	8	9	9
+1	10	2	3	4	5	6	7	8	9	9
+1	1	2	3	4	5	6	7	8	9	9
0	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
-1	0	0	1	2	3	4	5	6	7	8

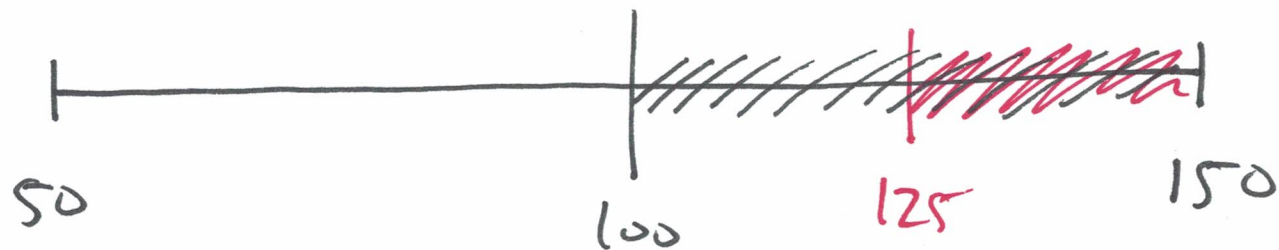
$$P(\text{FINAL SCORE} = 9) = \frac{8}{60} = \frac{2}{15}$$

$$P(\text{INITIAL SCORE} = 9) = \frac{5}{60}$$

$$P(\text{BOTH INITIAL AND FINAL} = 9) = \frac{5}{60}$$

$$P(\text{INIT} = 9 \mid \text{FINAL} = 9) = \frac{\frac{5}{60}}{\frac{8}{60}} = \frac{5}{8}$$

AGE OF TORTOISE



CONTINUOUS; UNIFORM DISTRIBUTION ON $[50, 150]$

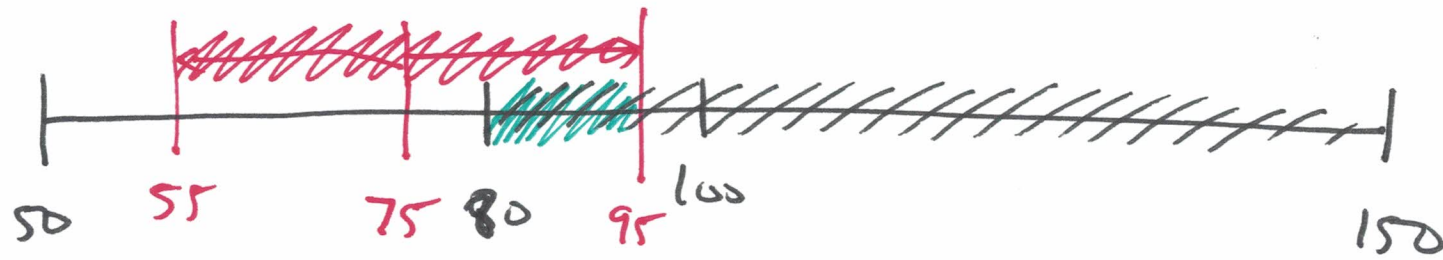
$$P(X > 125 \mid X > 100) = \frac{P(\text{Both})}{P(X > 100)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

$$P(X > 100) = \frac{1}{2}$$

$$P(X > 125 \text{ AND } X > 100) = P(X > 125) = \frac{1}{4}$$

Ed

$$P(|X-75| < 20 \mid X > 80)$$



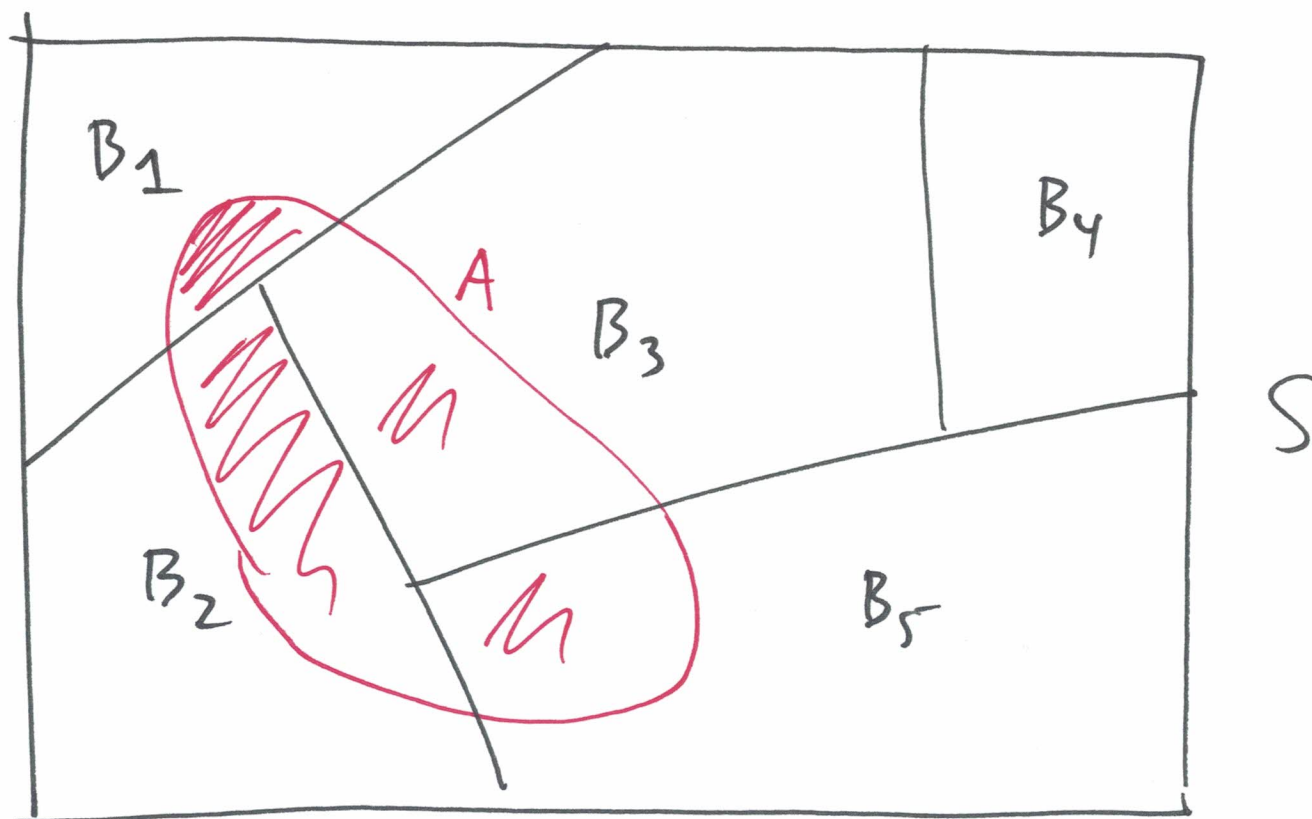
$$P(X > 80) = \frac{7}{10}$$

$$P(|X-75| < 20) = \frac{4}{10}$$

$$P(|X-75| < 20 \cap X > 80) = \frac{15}{100}$$

$$P(|X-75| < 20 \mid X > 80) = \frac{\frac{15}{100}}{\frac{70}{100}} = \frac{15}{70} = \frac{3}{14}$$

PARTITION OF SAMPLE SPACE:



$$\bigcup_{i=1}^{\infty} B_i = S \quad \text{AND}$$

$$B_i \cap B_j = \emptyset \quad \text{FOR ALL } i \neq j$$

~~$B_i \cap B_j = \emptyset$~~

SUBSETS COMPRISE ALL OF S

NO INTERSECTION

LET $\{B_i\}$ BE A PARTITION OF S .

$$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i)$$

$$= \sum_{i=1}^{\infty} P(A|B_i) \times P(B_i)$$

↳ BECAUSE ~~$P(A|B)$~~

$$P(A|B_i) = \frac{P(A \cap B_i)}{P(B_i)}$$

LAW OF TOTAL PROBABILITY.

EX AT A CONFERENCE,

50% RPI STUDENTS

30% MIT

20% HARVARD.

$$P(\text{ENG} \mid \text{RPI}) = 0.8$$

$$P(\text{ENG} \mid \text{MIT}) = 0.6$$

$$P(\text{ENG} \mid \text{HARVARD}) = 0.1$$

0.6.

"

$$\begin{aligned} P(\text{ENGINEER}) = & P(\text{ENG} \mid \text{RPI}) P(\text{RPI}) = (0.8)(0.5) \\ & + P(\text{ENG} \mid \text{MIT}) P(\text{MIT}) + (0.6)(0.3) \\ & + P(\text{ENG} \mid \text{H}) P(\text{H}) + (0.2)(0.1) \end{aligned}$$

BUT WHAT IS $P(B_i | A)$?

WE NEED BAYES' RULE:

$$P(B_i | A) = \frac{P(A \cap B_i)}{P(A)}$$

$$= \frac{P(A | B_i) P(B_i)}{P(A)}$$

$$= \frac{P(\text{ENG} | \text{RPI}) P(\text{RPI})}{P(\text{ENGINEER})} = \frac{0.4}{0.6} = \frac{2}{3}$$

KEY IDEA: PRIOR AND POSTERIOR
PROBABILITIES:

$P(B_i)$ = PRIOR PROBABILITY OF
CLASS.

$P(B_i | A)$ = POSTERIOR PROBABILITY.

IN EXAMPLE: 50% $\rightarrow$ 67%

NOTE

$$\frac{P(A|B_i) P(B_i)}{P(A)}$$

$$= \frac{P(A|B_i) P(B_i)}{\sum_{i=1}^{\infty} P(A|B_i) P(B_i)}$$

CONSIDER TESTING FOR FLU, COVID

~~WILL~~ DISEASE IS

HERE'S OUR TEST:

$$P(T \text{ POSITIVE} \mid \text{DISEASE}) = 0.73$$

SENSITIVITY

$$P(T \text{ NEGATIVE} \mid \text{DISEASE}) = 0.27$$

FALSE
NEGATIVE

$$P(T \text{ POSITIVE} \mid \text{NO DISEASE}) = 0.0004$$

SPECIFICITY

FALSE
POSITIVE

$$P(T \text{ NEGATIVE} \mid \text{NO DISEASE}) = 0.9996$$

WE ALSO HAVE THE
PRIOR PROBABILITIES

$$\begin{array}{l} P(\text{DISEASE}) \\ P(\text{NO DISEASE}) \end{array}$$

WHAT IF T IS POSITIVE? DO WE
HAVE THE DISEASE?

$$P(\text{DISEASE} \mid T \text{ POS}) = \frac{\cancel{P(T \text{ POS} \mid D)} \cancel{P(D)}}{\cancel{P(T \text{ POS})} P(T \text{ POS})}$$

$$\frac{P(T \text{ POS} \mid D) P(D)}{P(T \text{ POS})}$$

$$= \frac{P(T \text{ POS} \mid D) P(D)}{P(T \text{ POS} \mid D) P(D) + P(T \text{ POS} \mid \text{NO } D) P(\text{NO } D)}$$

$$P(T \text{ POS} \mid D) P(D) + P(T \text{ POS} \mid \text{NO } D) P(\text{NO } D)$$

SUPPOSE THAT $P(D) = 0.001$

$$\frac{(0.73)(0.001)}{(0.73)(0.001) + (0.0004)(0.999)} = 0.65.$$

SEEMS LOW!

$$\text{IF } P(\text{DISEASE}) = 0.01 \Rightarrow P(D|T \text{ pos}) = 0.95$$

$$P(\text{DISEASE}) = 0.1 \Rightarrow P(D|T \text{ pos}) = 0.995$$